Module & global coordinates

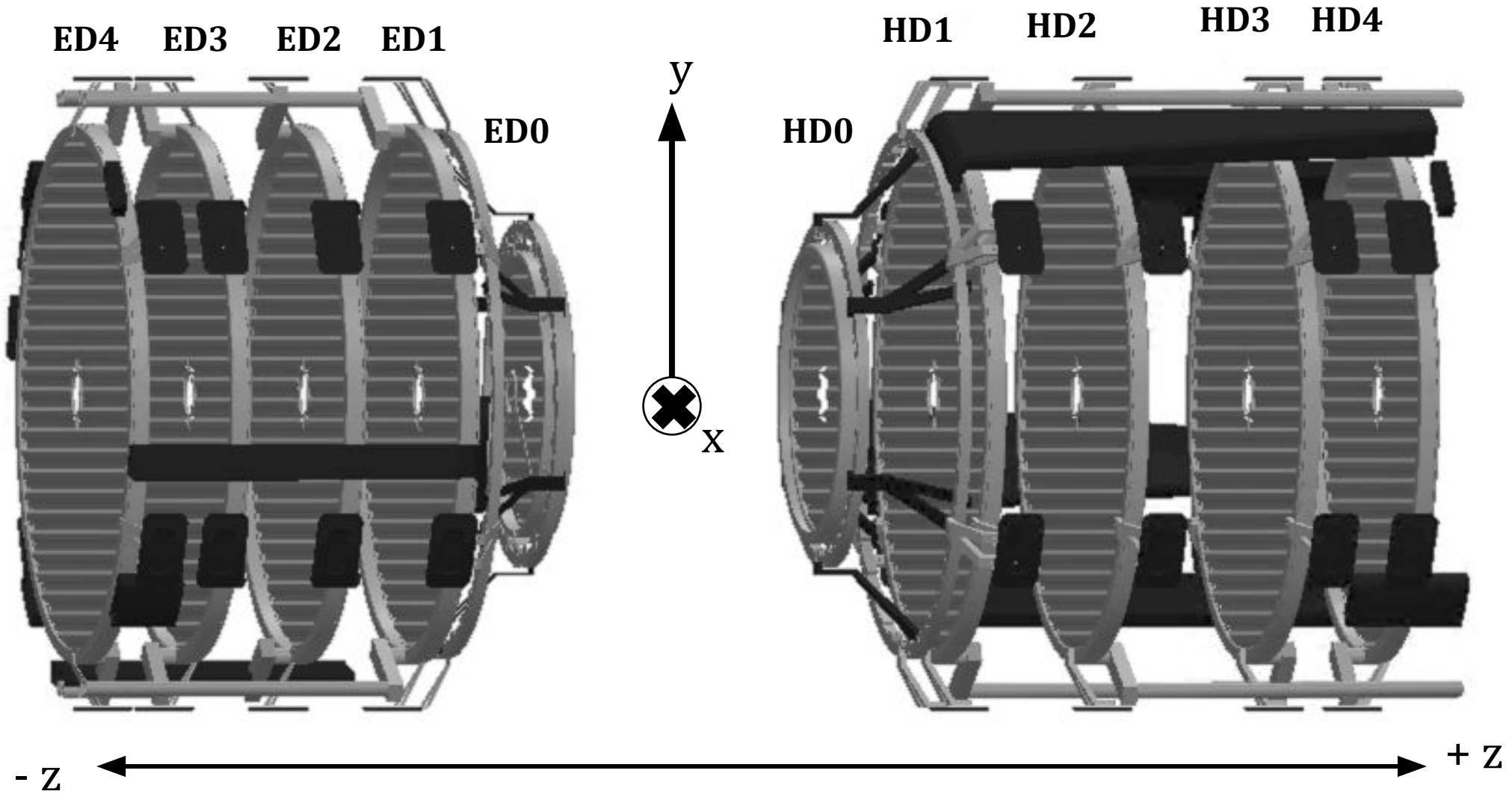
Author: Nicole Apadula

Version: 1.0

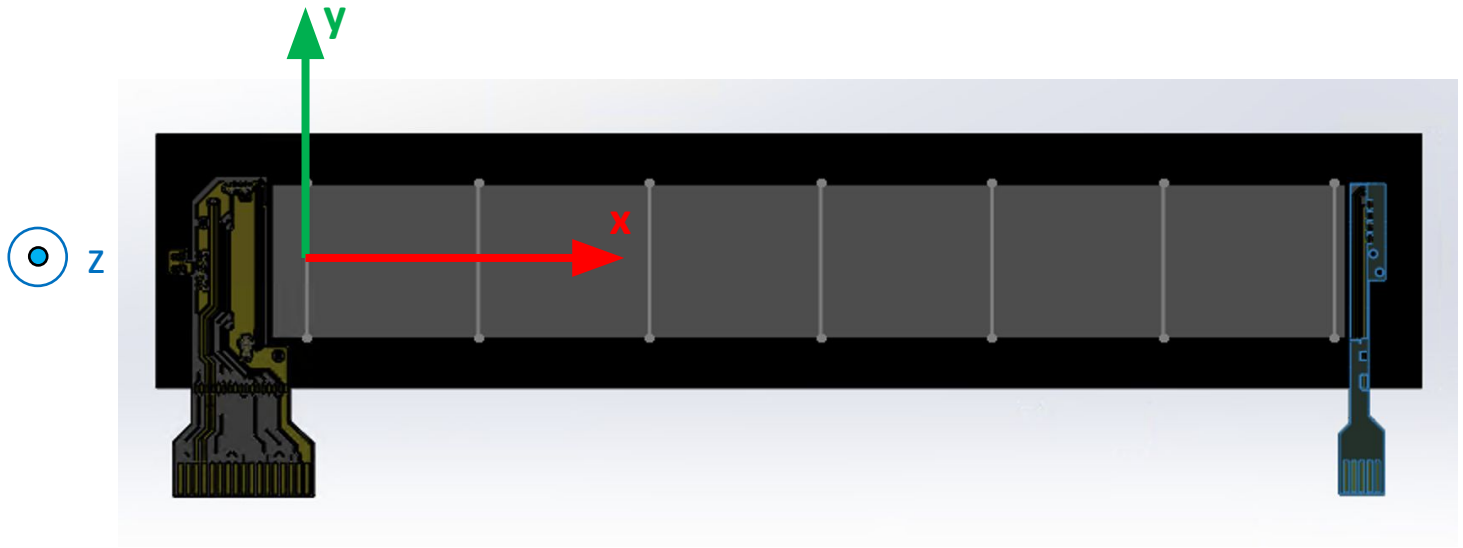
Date: 10/16/2026

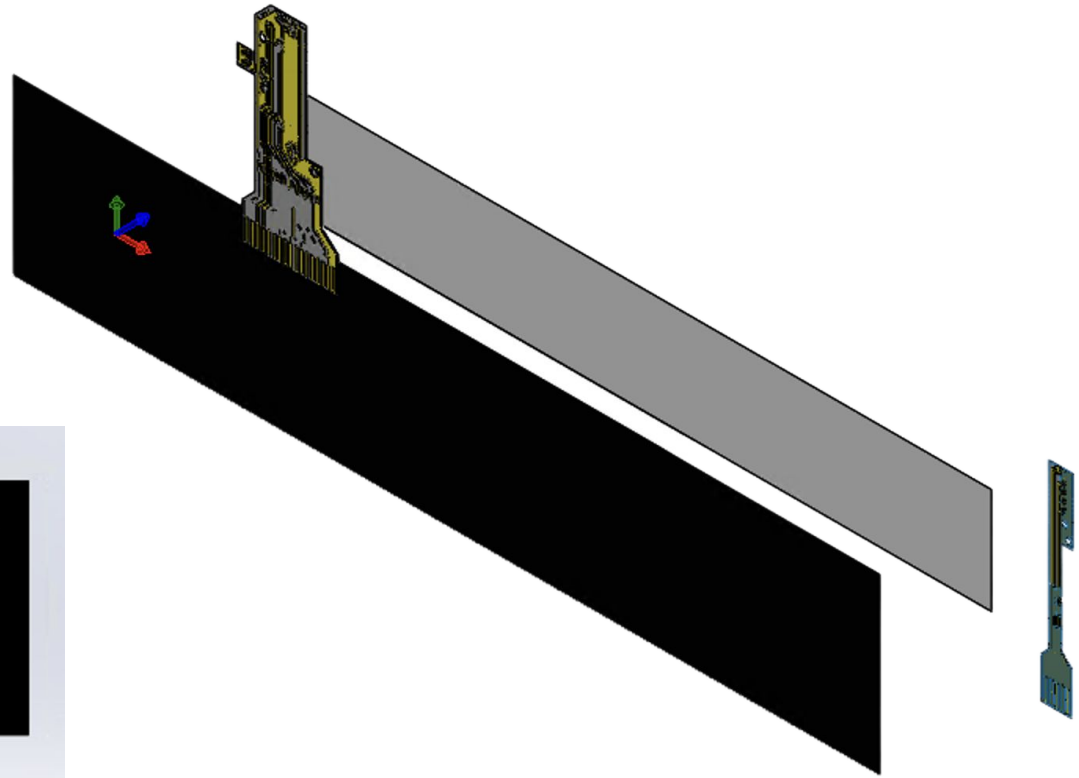
Definitions of local and global coordinates and mapping onto disks

Right-handed global coordinate system



Local module coordinates





Local coordinate system is defined by the sensor. Origin is at the border between LEC & RSUs
X goes towards REC
Y goes upwards towards the top tiles from the center of the RSU
Z comes out away from the carbon fiber

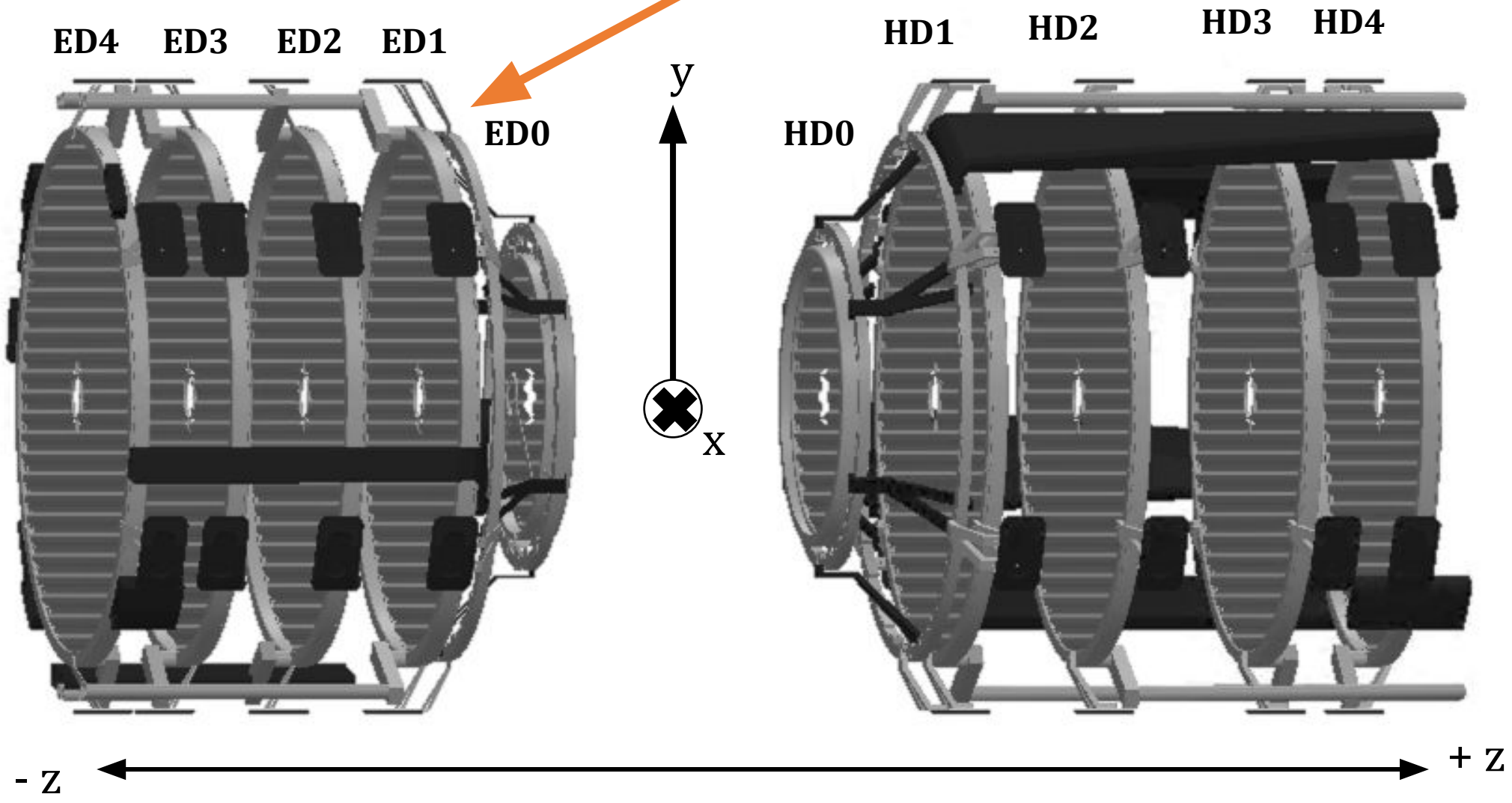
Local module coordinates

Direction of FPC does not change local coordinate system

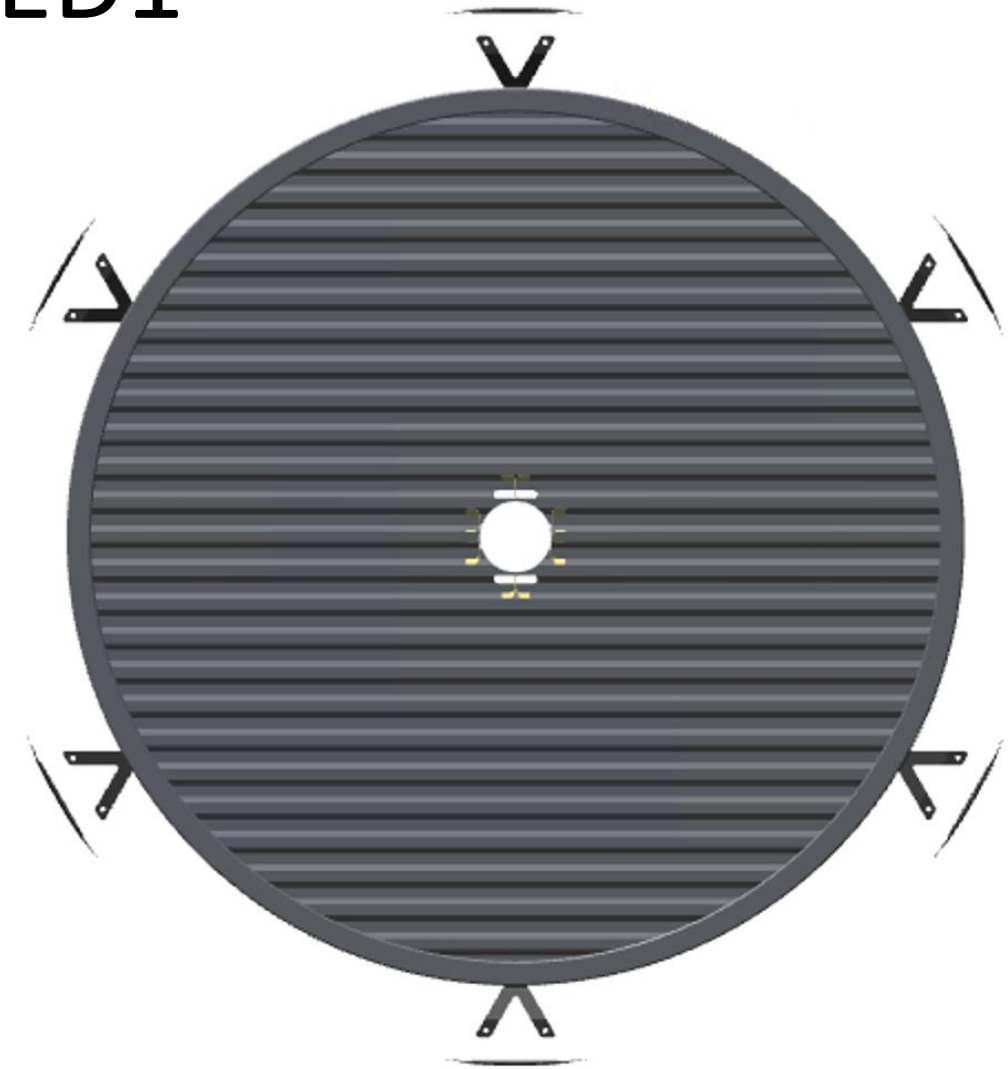


Global coordinates

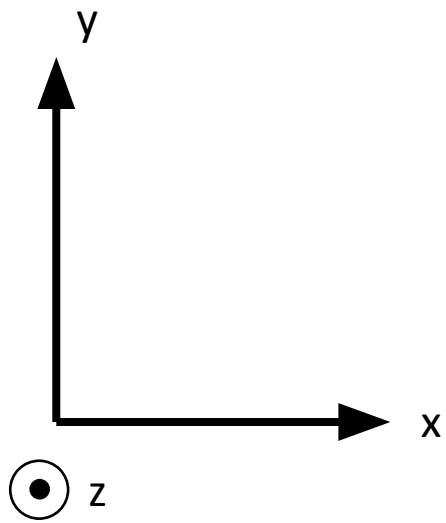
Picking one case: ED1



One side of ED1

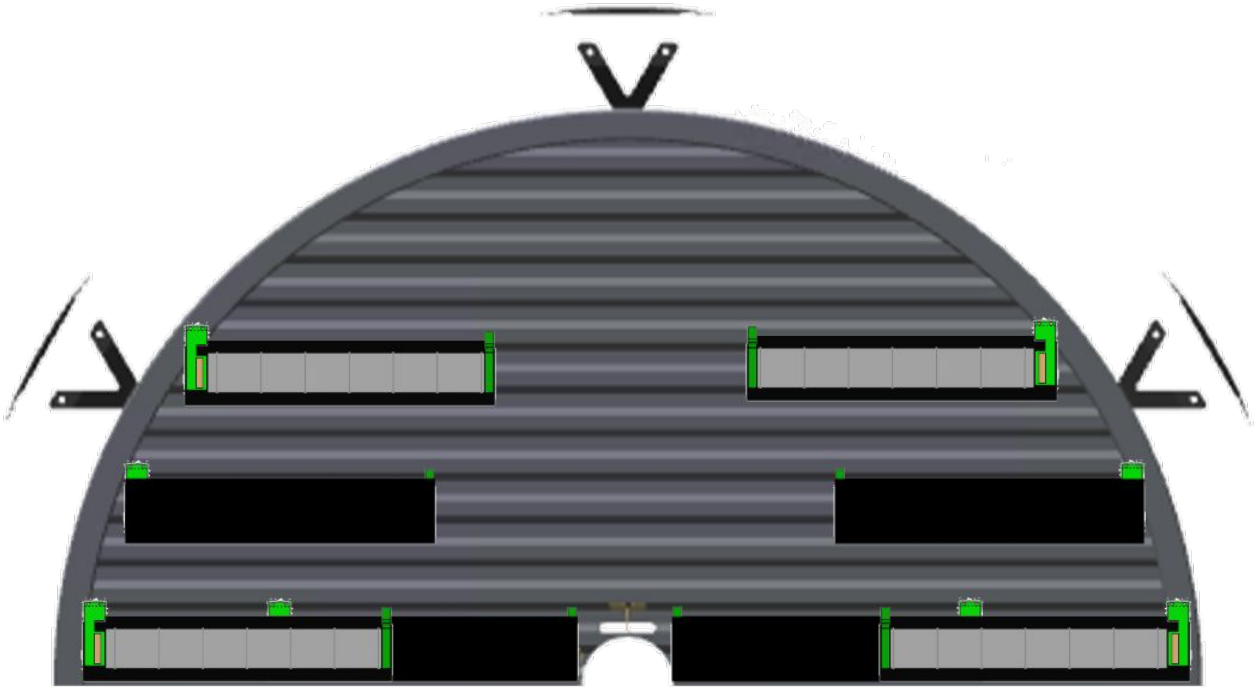


Global coordinates

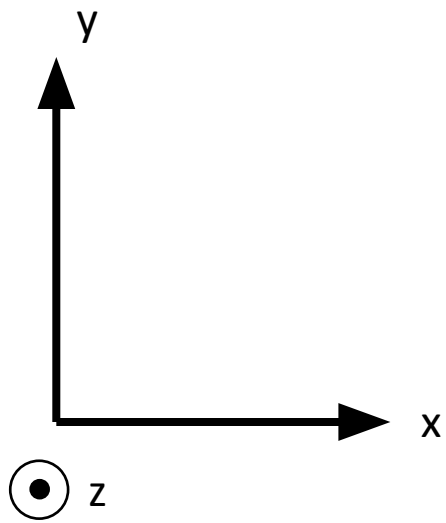


Modules used on ED1

Only taking this one case scenario:
Top half (defined by being in pos global y)
Back face (defined by being at larger global z)

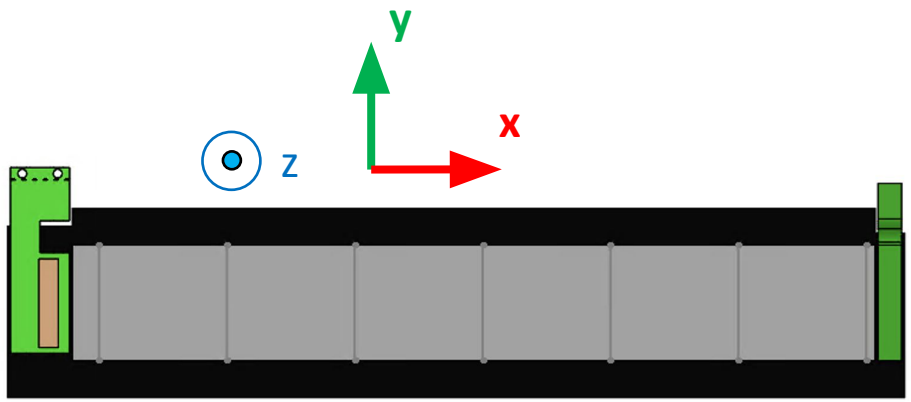


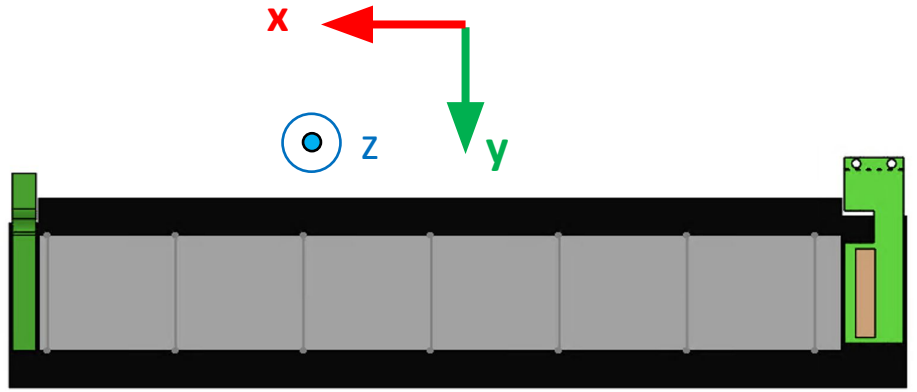
Global coordinates

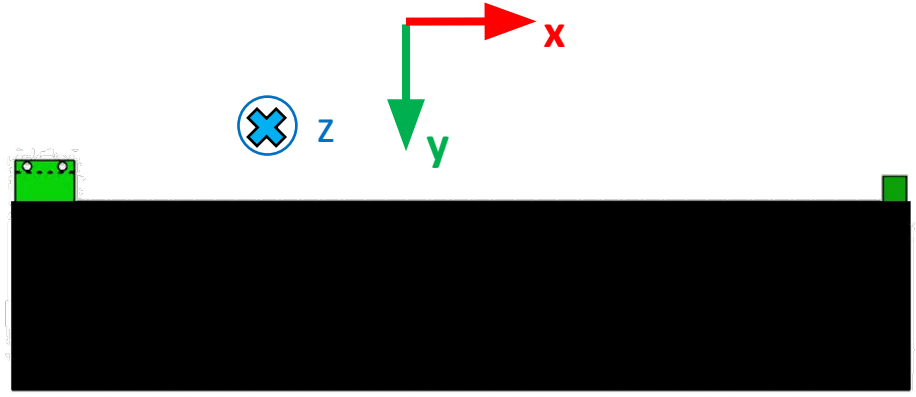


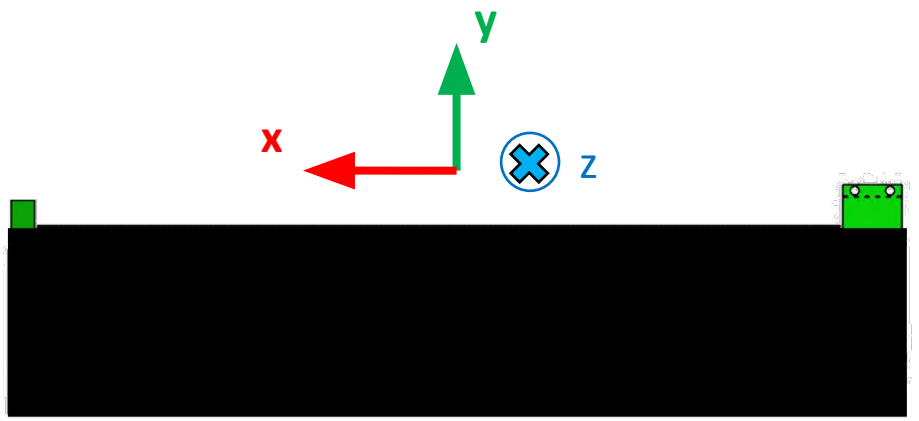
4 modules orientations used, close ups shown in the next slide

Local module coordinates – module rotations on ED1 case (previous slide)

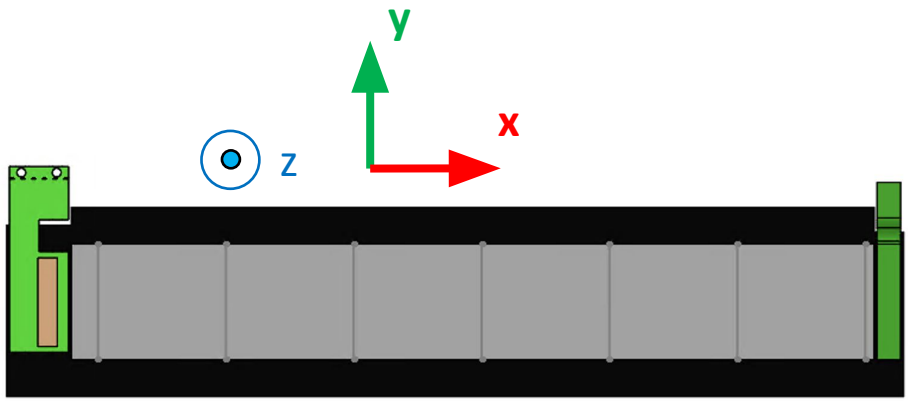




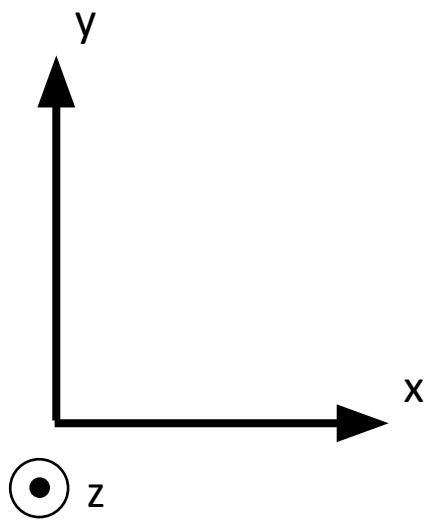




Mapping to global coordinates



Global coordinates

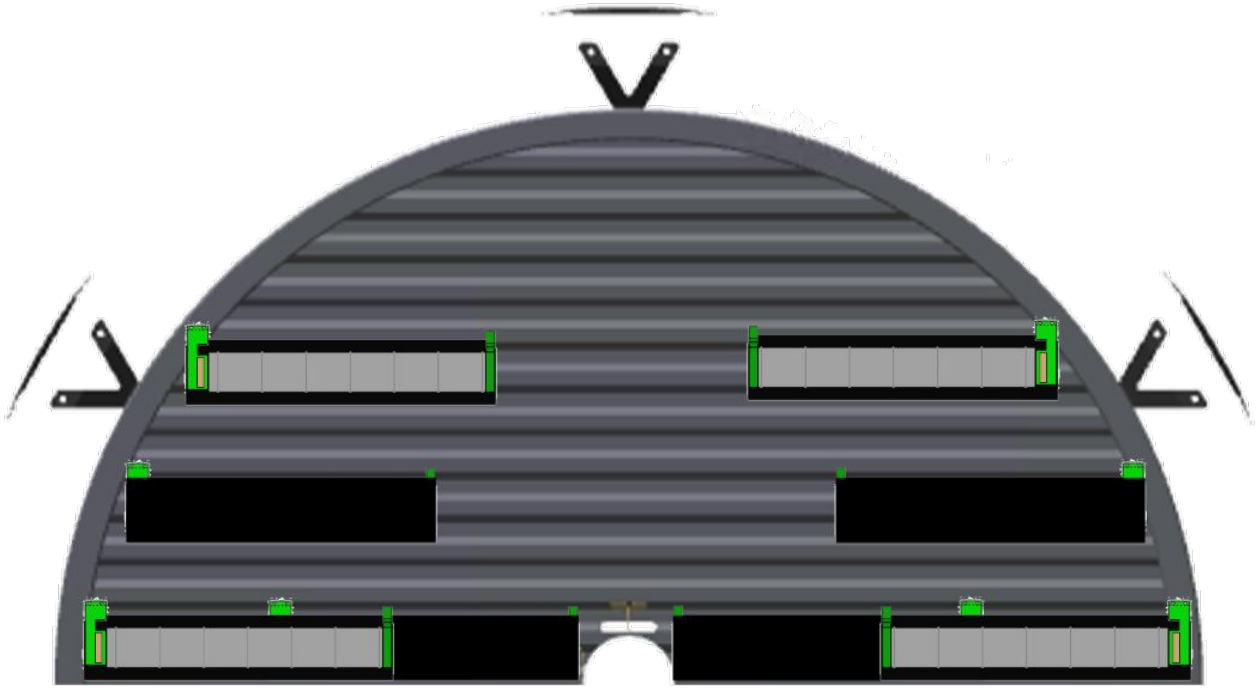


sx = 1

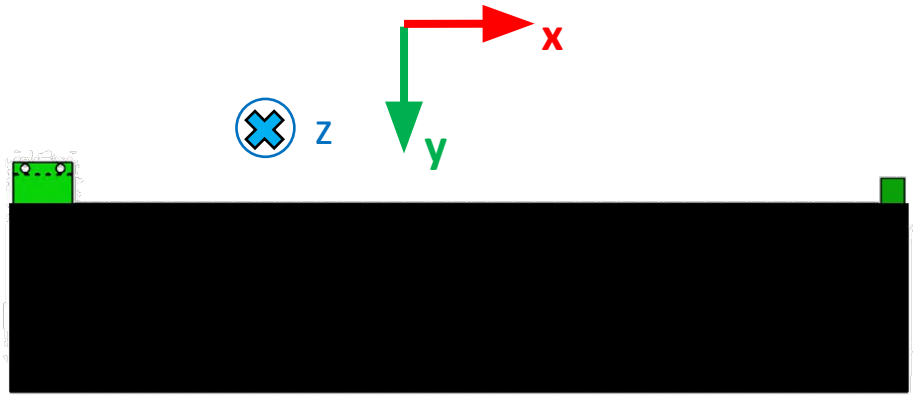
sy = 1

sz = 1

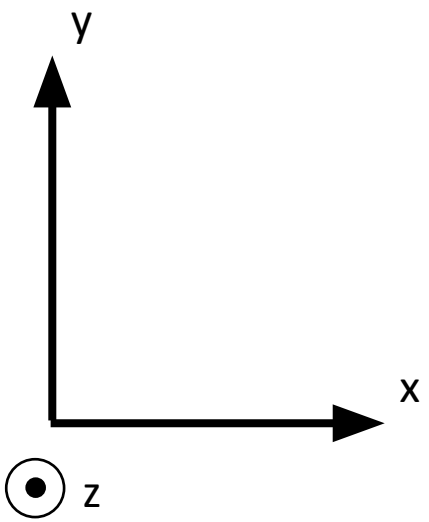
⟶ sx*sy*sz = 1, Left Chirality



Mapping to global coordinates



Global coordinates

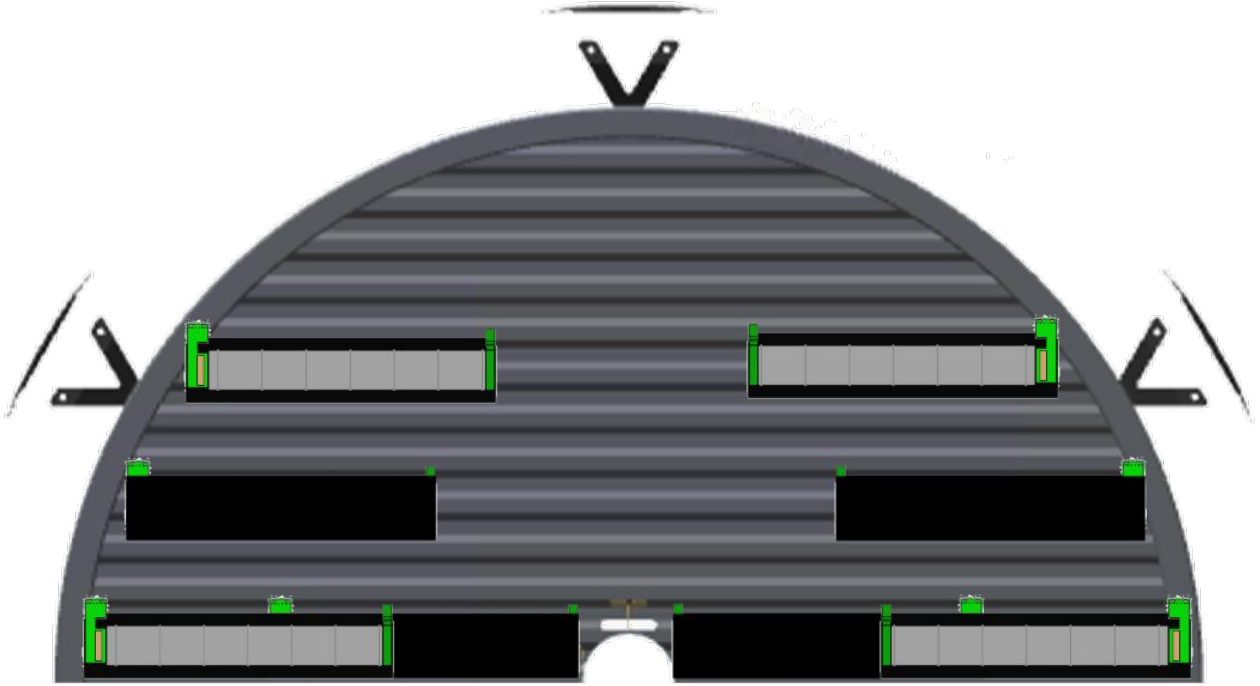


sx = 1

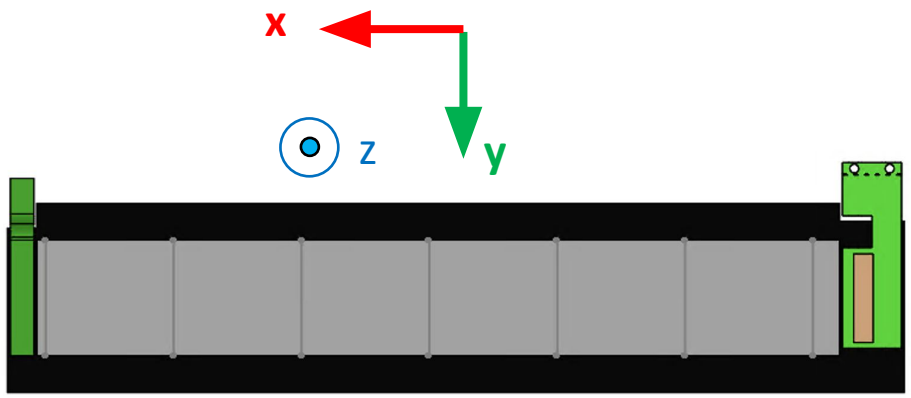
sy = -1

sz = -1

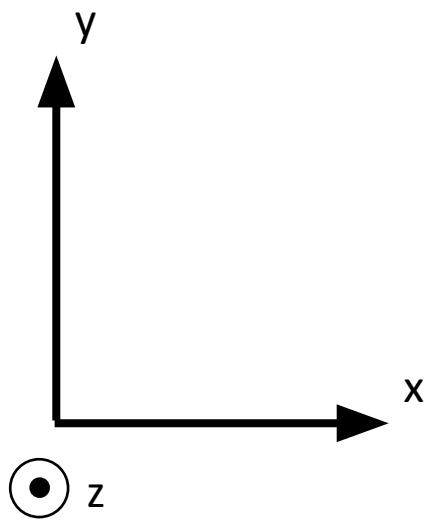
⟶ sx*sy*sz = 1, Left Chirality



Mapping to global coordinates



Global coordinates

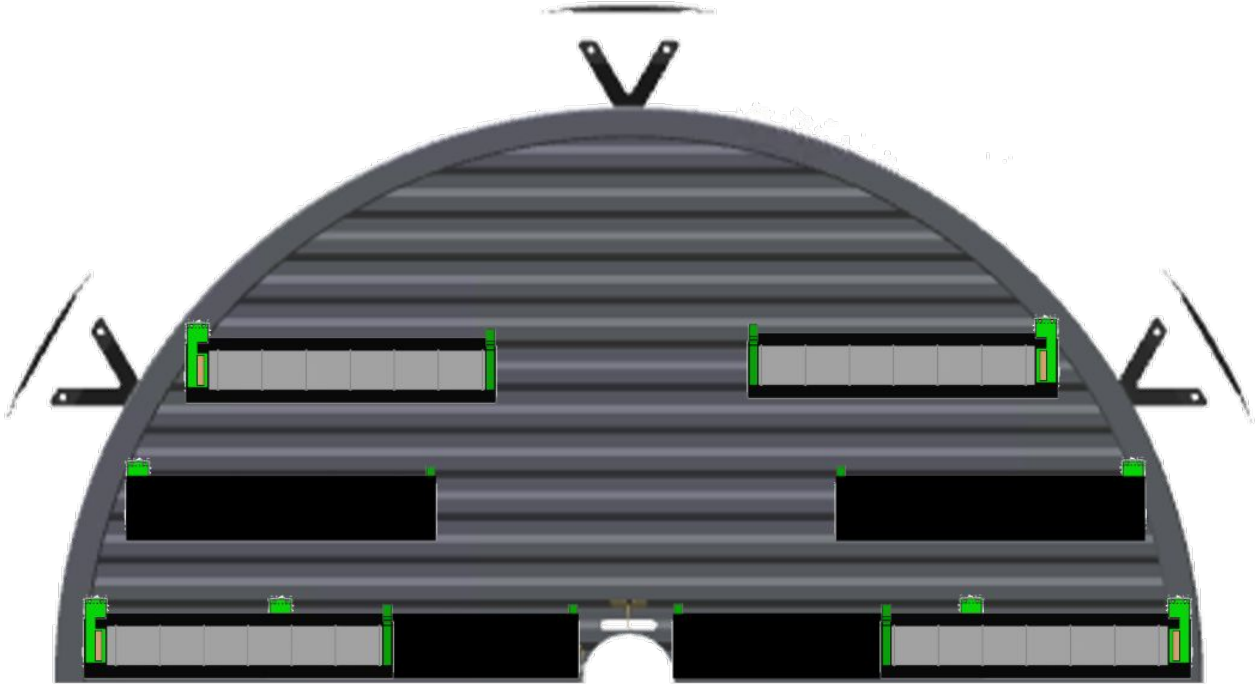


sx = -1

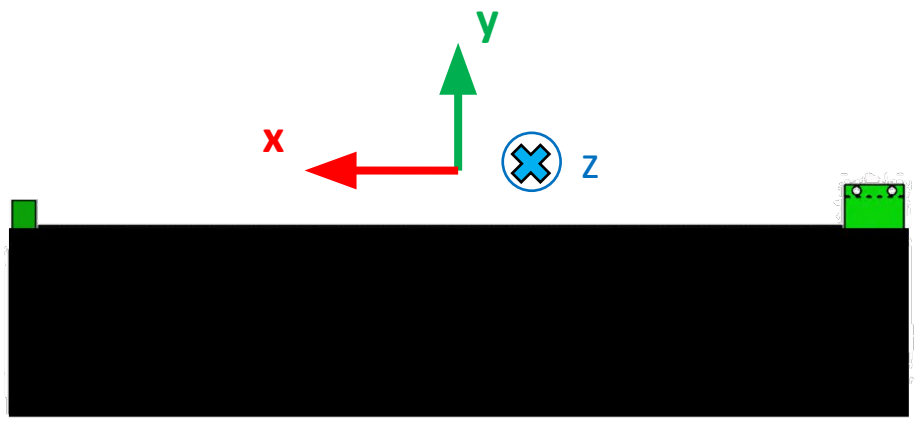
sy = -1

sz = 1

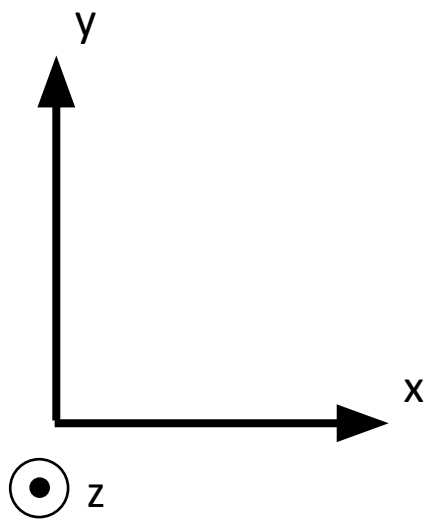
→ sx*sy*sz = 1, Left Chirality



Mapping to global coordinates



Global coordinates

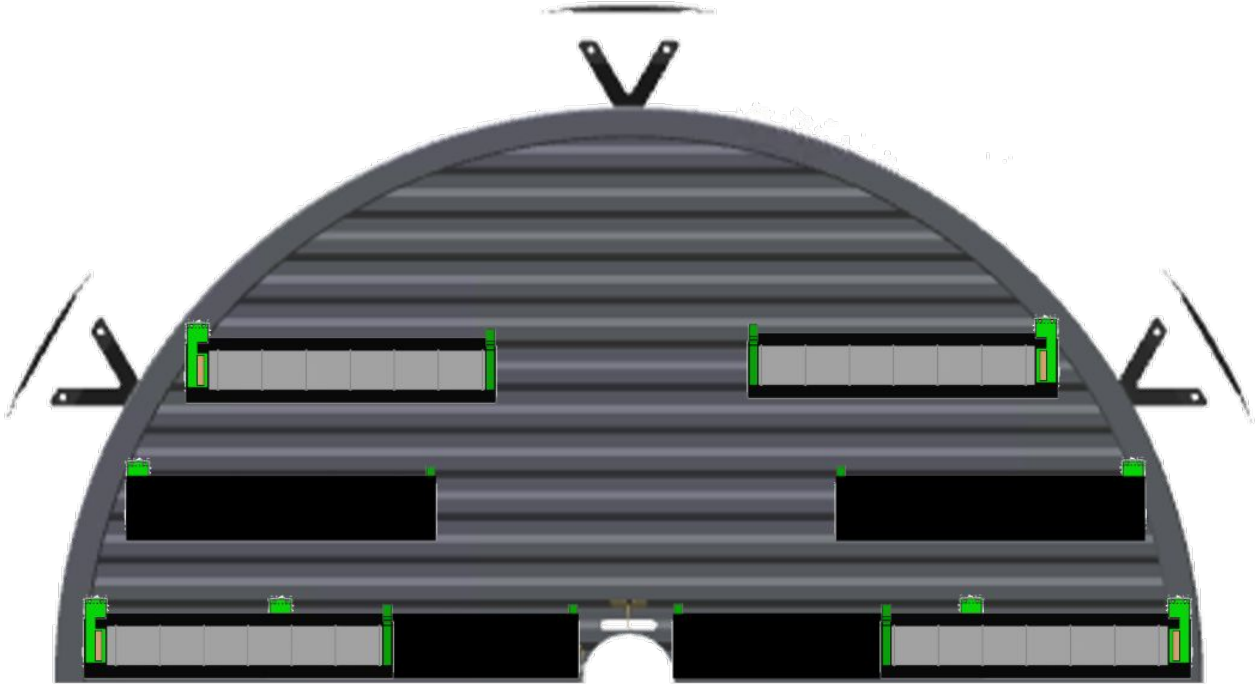


sx = -1

sy = 1

sz = -1

→ sx*sy*sz = 1, Left Chirality



Proposition

- Modules are defined locally
- All movements are based on the currently defined module coordinate system and the local origin (defined earlier)
- Direction of FPC defines handedness
  - The FPC either extends to positive y (right handed) or to negative y (left handed)
- Inward or outward facing flips the z
  - Inward and the z axis is now extending away locally

WIP for naming